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Artificial Intelligence and Learning Gaps: Evaluating the Effectiveness of Personalized Pathways

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Abstract

The integration of Generative AI (GAI) in education has opened new possibilities for personalized learning, yet its effectiveness in mitigating learning gaps remains underexplored. This study examines the impact of Personalized Learning Pathways (PLPs), generated through AI models (Gemini 2.5 Pro, ChatGPT 5), on secondary school students' learning outcomes. Using a short-term longitudinal panel design, the research compares homogeneous instructional strategies with AI-driven personalized learning to assess differences in knowledge acquisition and cognitive skill development. Findings indicate that AI-generated PLPs significantly reduce lower-order learning gaps, though higher-order skills remain challenging. The study also reveals that learning styles influence student engagement with AI-driven education, suggesting that hybrid models combining AI and teacher mediation may optimize outcomes. These findings contribute to the ongoing discourse on AI in education, emphasizing the need for equitable, adaptive, and ethical AI applications in learning environments.

Keywords: personalized learning pathways; artificial intelligence; learning gaps; learning styles; education 4.0

1. Introduction

In the era of Artificial Intelligence (AI), education is undergoing a transformation that requires institutions and teachers to adapt to new learning dynamics. In this context, Education 4.0 promotes the integration of emerging technologies to respond to diverse learner needs and to enable more flexible forms of instruction [1]. Accordingly, AI has been positioned as a driver of adaptive learning environments that can enrich learning experiences [2,3]. Nevertheless, the educational promise of these technologies remains constrained by persistent challenges, particularly the difficulty of addressing learning gaps that accumulate over time.

These learning gaps, understood as deficiencies in essential knowledge or skills, tend to widen when they are not addressed early and systematically. This risk is especially pronounced among students from vulnerable backgrounds, who often have limited access to academic support mechanisms beyond the classroom [4]. As a consequence, gaps can undermine academic continuity and contribute to frustration and disengagement, thereby reinforcing inequities in educational trajectories. Against this backdrop, AI-enabled personalisation has been proposed as a means to detect and respond to emerging gaps through automated and adaptive support.

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However, homogeneous teaching strategies often struggle to meet the needs of increasingly diverse classrooms. In this regard, Akpan [5] and Zhang et al. [6] highlight that conventional instruction tends to benefit only a portion of students, while others risk incomplete learning or even disengagement from schooling. Therefore, personalised learning has been positioned as a promising alternative because it can adjust content, pacing, and instructional approaches to individual profiles [7]. At the same time, its large-scale implementation remains hindered by barriers such as infrastructure constraints and gaps in teacher preparation [8], which points to the need for feasible and classroom-compatible personalisation strategies.

Within this landscape, Personalised Learning Pathways (PLPs) have emerged as a relevant strategy because they structure individualised learning sequences that allow students to progress at an appropriate pace and according to their specific needs [9]. Moreover, PLPs can incorporate varied resources, such as readings, simulations, and videos, thus expanding instructional opportunities and supporting diverse forms of engagement. In turn, generative AI can strengthen PLP design by automating the curation and generation of tailored learning resources and prompts [10]. Consequently, tools such as ChatGPT 5 and Gemini 2.5 Pro can support dynamic adjustments to learning pathways in near real time [11,12], which makes PLPs particularly salient for contexts where teachers need scalable ways to respond to heterogeneous learning needs.

Despite these advances, the evidence base on AI-enabled personalisation and PLP design remains limited in ways that constrain actionable educational guidance. Notably, much of the existing literature emphasises AI as a means to automate teaching-related processes, while offering less clarity about how such approaches influence knowledge acquisition and the reduction in specific learning gaps [13,14]. In parallel, the role of learning styles continues to be contested, with some authors defending their instructional relevance [15] and others questioning their scientific validity [16]. Accordingly, there remains a significant research gap in studies that jointly examine generative-AI-supported PLPs, learning-style criteria, and learning-gap mitigation, as summarised in Figure 1.

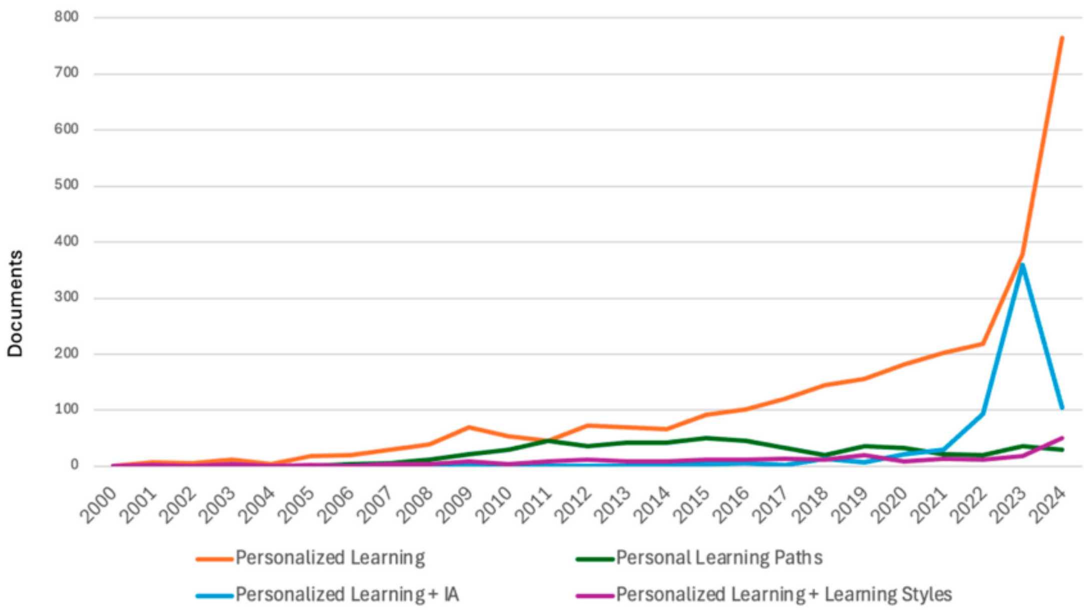

Figure 1. Research on PL, LS, and PLPs.

To address this gap, this paper makes three contributions. First, it reports evidence from a 24-week, cycle-based implementation with 21 eighth-grade students in a low-income public-school context, contrasting a homogeneous whole-class approach with generative-

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AI–supported Personalised Learning Pathways (PLPs). Second, it operationalises learning gaps using Bloom-aligned formative assessments and traces change across three monthly cycles, combining qualitative analysis of student responses and field diaries with quantitative summaries by cognitive level. Third, it examines how learning-style preferences relate to students' uptake of AI-generated explanations under varying levels of relevant cognitive load, and it derives design and equity implications for classroom-feasible, hybrid human-AI personalisation.

Against this background, this study examines the cyclical implementation of PLPs constructed with the support of generative AI in secondary education, incorporating learning styles as a personalisation criterion. Specifically, it evaluates the extent to which these pathways mitigate learning gaps and explores equity-related implications by considering students with different socioeconomic profiles and levels of prior competence. In this regard, the next section presents the research question that guided the study.

2. Research Questions

From this perspective, the research question arises: To what extent are personalized learning pathways, constructed based on learning styles and generative artificial intelligence, effective in overcoming learning gaps identified in secondary school students?

This research approach aims to contribute to the debate on personalized learning and the role of AI in education, exploring innovative solutions that intertwine technology and pedagogy to create more inclusive, adaptive, and sustainable learning environments.

3. Literature Review

3.1. Education in the Age of Artificial Intelligence

Artificial Intelligence has moved from being a peripheral educational technology to becoming part of the infrastructure through which learning is organised, monitored, and increasingly shaped. Within Education 4.0, this shift is often framed as a transition towards data-intensive, adaptive, and more efficient learning ecosystems, where analytics and automation enable faster instructional decisions and more responsive learning environments [2,17–19]. Nevertheless, the literature also suggests that AI in education remains an evolving field in which the distance between technical potential and pedagogical impact is still substantial, particularly when equity and real-world classroom constraints are considered [20]. Consequently, rather than assuming AI-driven improvement as a given, current scholarship increasingly calls for empirical examinations that make explicit what "improvement" means, for whom it occurs, and under what conditions.

3.2. The Relevance of Personalisation in Contemporary Education

Against this backdrop, personalisation has gained prominence as a pedagogical response to heterogeneity, challenging the adequacy of one-size-fits-all instruction for learners who differ not only in prior knowledge and goals but also in affective, sociocultural, and behavioural dimensions [7,21–23]. AI has strengthened this agenda by enabling faster cycles of feedback, adaptive support, and data-informed adjustments, while also amplifying debates about what is being personalised: learning pathways, resources, pacing, assessment, or the learner experience itself [24,25]. Importantly, historical precedents, from Pressey's teaching machines to Skinner's programmed instruction, show that the promise of individualised support is not new; what is new is the scale and speed at which algorithms can now generate, adapt, and distribute learning experiences [26]. However, the same historical lens also cautions that instructional automation does not automatically translate into better learning, and that pedagogical coherence must be demonstrated rather than presumed.

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3.3. Personalised Learning Pathways as a Mechanism for Educational Customisation

Personalised Learning Pathways (PLPs) have been positioned as a practical mechanism to operationalise personalisation by structuring modular trajectories that respond to learners' evolving needs and performance [27]. In principle, PLPs translate the ideal of adaptation into sequences of tasks and resources that are adjusted through ongoing evidence of learning, often supported by learning analytics and iterative feedback loops [28]. Yet, the literature also indicates that PLP effectiveness depends less on the mere existence of a pathway and more on the quality of its pedagogical design, as mentioned by Chou et al. [29] and Scheiter et al. [30]: the logic of progression, the nature of tasks, and the way guidance supports learners when cognitive demands increase. From this perspective, Generative AI introduces a qualitatively different affordance: it can move beyond recommending resources and begin to generate explanations, examples, activities, and alternative representations on demand, potentially reducing friction in pathway construction and enabling rapid personalisation at scale [31,32]. At the same time, such generativity raises questions about alignment, consistency, and instructional appropriateness, reinforcing the need for classroom-based evaluations that do not treat PLPs as a purely technical solution.

3.4. The Role of Learning Styles in Personalised Learning Pathways

The inclusion of learning styles in personalised learning remains controversial. On the one hand, some studies report that aligning resources with declared preferences can increase engagement and perceived usefulness, particularly in technology-mediated environments [15,33]. On the other hand, critics caution that rigid categorisations risk oversimplifying learning processes and may produce instructional decisions that are not grounded in robust evidence, especially when styles are treated as stable traits rather than context-sensitive preferences [16]. Within this debate, the Honey and Mumford model, derived from Kolb, offers a practical vocabulary (active, reflective, pragmatic, theoretical) that is frequently used in applied educational settings [15]. Recent implementations suggest that AI-based systems can incorporate such preferences dynamically, adjusting the form and sequence of resources rather than locking learners into a single pathway logic [34–37]. Nevertheless, evidence also supports a more nuanced stance: if learning styles are used, they should be treated as one cue among others, complemented by data-driven indicators of learning progress to avoid overfitting personalisation to a debatable construct [38].

3.5. Learning Gaps and the Need for Adaptive Learning Solutions

Learning gaps remain a central challenge because they are cumulative: deficiencies in foundational knowledge and skills tend to compound over time, often widening inequalities and limiting learners' capacity to access more complex content [39]. In this regard, Vargas Mantilla et al. [4] mention that this risk is accentuated in vulnerable contexts where access to support mechanisms is uneven and educational disruption is more frequent. The literature has therefore emphasised adaptive solutions that combine formative assessment with targeted instructional responses, enabling gaps to be identified early and addressed through timely, tailored interventions [5,40]. So, as mentioned by Iatrellis et al. [41] and Taherisadr et al. [32], generative AI has been framed as a potential accelerator of this cycle, as it can generate customised explanations, practice items, and feedback in response to emerging difficulties, rather than only recommending pre-existing materials. However, scholarship also raises concerns about the pedagogical and ethical implications of AI-mediated support, particularly the risk of reducing learner agency, encouraging over-reliance, and obscuring accountability for instructional decisions [42]. These tensions reinforce the importance of studying not only whether gaps diminish, but also how learners engage with AI support and what forms of guidance remain necessary.

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3.6. Integrating AI, Personalised Learning Pathways, and Learning Styles to Mitigate Learning Gaps

As mentioned so far, the literature suggests a plausible proposition: AI-generated PLPs may be particularly well suited to supporting gap reduction when they combine timely diagnosis with targeted, adaptive practice and feedback, while maintaining pedagogical coherence and appropriate scaffolding [9,12]. At the same time, it is not evident that the same mechanisms that help consolidate lower-order learning will, on their own, produce consistent gains in higher-order reasoning, especially when cognitive demands increase and when learners require social mediation and structured challenge [43,44]. Moreover, the role of learning styles within this integration remains unresolved: while preference-based adjustments may improve engagement for some learners, the construct itself is contested and may require cautious, flexible use rather than deterministic pathway design [16,38].

Consequently, a key gap persists in empirical evidence that simultaneously examines AI-generated PLPs as an instructional mechanism, learning gaps as an outcome assessed through iterative formative cycles, and learner variability, approached cautiously through learning styles, within authentic secondary-school contexts characterised by equity constraints. Addressing this gap requires classroom-based studies that compare homogeneous instruction with AI-supported PLPs, documenting not only outcomes but also the conditions under which personalisation functions as genuine pedagogical support rather than as a technological promise.

4. Method

4.1. Study Design

A qualitative study with a non-experimental short-term longitudinal panel design was implemented [45–47] to evaluate the effectiveness of personalized learning pathways (PLPs) based on generative AI (GAI) in mitigating learning gaps in secondary school students. This design allowed for tracking knowledge acquisition and learning gap reduction over a short timeframe, without manipulating variables.

Unlike experimental designs, a non-experimental short-term longitudinal panel approach enables natural observation of learning changes at multiple points, making it suitable for educational research constrained by ethical and practical limitations [46]. By repeatedly assessing the same individuals, the study identifies patterns of improvement, stagnation, or decline in learning, offering insights into how AI-generated PLPs support knowledge retention and academic progress. This methodology aligns with the study's objective of determining whether AI-generated PLPs significantly reduce learning gaps. It also accounts for factors such as prior knowledge, learning styles, and technology accessibility, overcoming the limitations of cross-sectional designs, which provide only static observations instead of progressive learning data.

This design was selected because the research question required tracing within-student learning trajectories across repeated classroom cycles under authentic constraints, rather than estimating a single end-point effect under controlled conditions. A non-experimental short-term longitudinal panel approach is suitable in school settings where random assignment and strict manipulation are impractical, while still enabling systematic observation of change over time. To strengthen credibility, the study relied on multiple sources of evidence (formative assessment artefacts, field diaries, and student-generated outputs), and it prioritised transparency in how learning gaps were identified and categorised across cycles. Accordingly, the goal of the design is analytic generalisation, clarifying mechanisms and boundary conditions for GenAI-supported PLPs, rather than statistical generalisation to broader populations.

4.2. Participants and Context

The study sample consisted of 21 eighth-grade students from a public school at Bogotá, Colombia. Of these, 12 were female and 9 were male, with ages ranging from 11 to 15 years. The group included 14 local and 7 migrant students, reflecting the diversity present in urban educational contexts with migrant populations. Socioeconomic characteristics revealed that all participants belonged to low-income context. Only 15 students had full access to a technological device for school activities, while 4 had to share a device and 2 lacked any device at home. Regarding internet access, 9 had Wi-Fi at home, 5 relied on daily mobile data top-ups, and 5 used monthly top-ups, highlighting potential challenges in accessing educational and technological resources. To ensure voluntary participation and ethical compliance, parental or guardian written informed consent was obtained. Four students were excluded due to a lack of legal authorization.

The sample size reflects a classroom-bounded cohort and an information-rich case, enabling dense observation across multiple measurement points and instructional cycles. Because the study follows the same learners longitudinally and triangulates multiple data sources, the emphasis is placed on the depth and coherence of evidence rather than on representativeness. Therefore, findings are interpreted as analytically transferable to contexts with comparable conditions (e.g., secondary education, heterogeneous prior knowledge, and equity constraints), while acknowledging that broader population-level claims would require larger, multi-site designs.

4.3. Procedure

The study was conducted for 24 weeks, structured into three major stages, in which several processes were executed, as shown in Figure 2.

4.3.1. Setup Stage

The setup stage established the foundation for the study by defining the instructional framework, ensuring the reliability of data collection tools, and implementing methodological strategies to enhance external consistency. As will be detailed later, this phase included curricular planning, where structured learning sequences were designed to align with cognitive learning theories. Additionally, instrument design and validation ensured the accuracy of assessment tools, reinforcing internal consistency. Lastly, strategies for improving transferability and external reliability were integrated to enhance the applicability of findings across similar educational settings, including detailed contextual descriptions, triangulation methods, and expert judgment evaluations.

Curricular Planning

The instructional design was based on Ausubel's cognitive progression principles to ensure structured and sequential learning and Reigeluth's elaboration theory, which starts with a simplified general overview before delving into more complex concepts. Additionally, Sweller′s cognitive load theory was applied to optimize information processing and knowledge retention. The learning units aligned with Colombia′s eight grade curriculum and the United Nations′ Sustainable Development Goal 7 (Affordable and clean energy). In addition, Bloom's taxonomy guided the progression of cognitive skills from basic (knowledge, comprehension, application) to higher-order skills (analysis, evaluation, synthesis) [5,48].

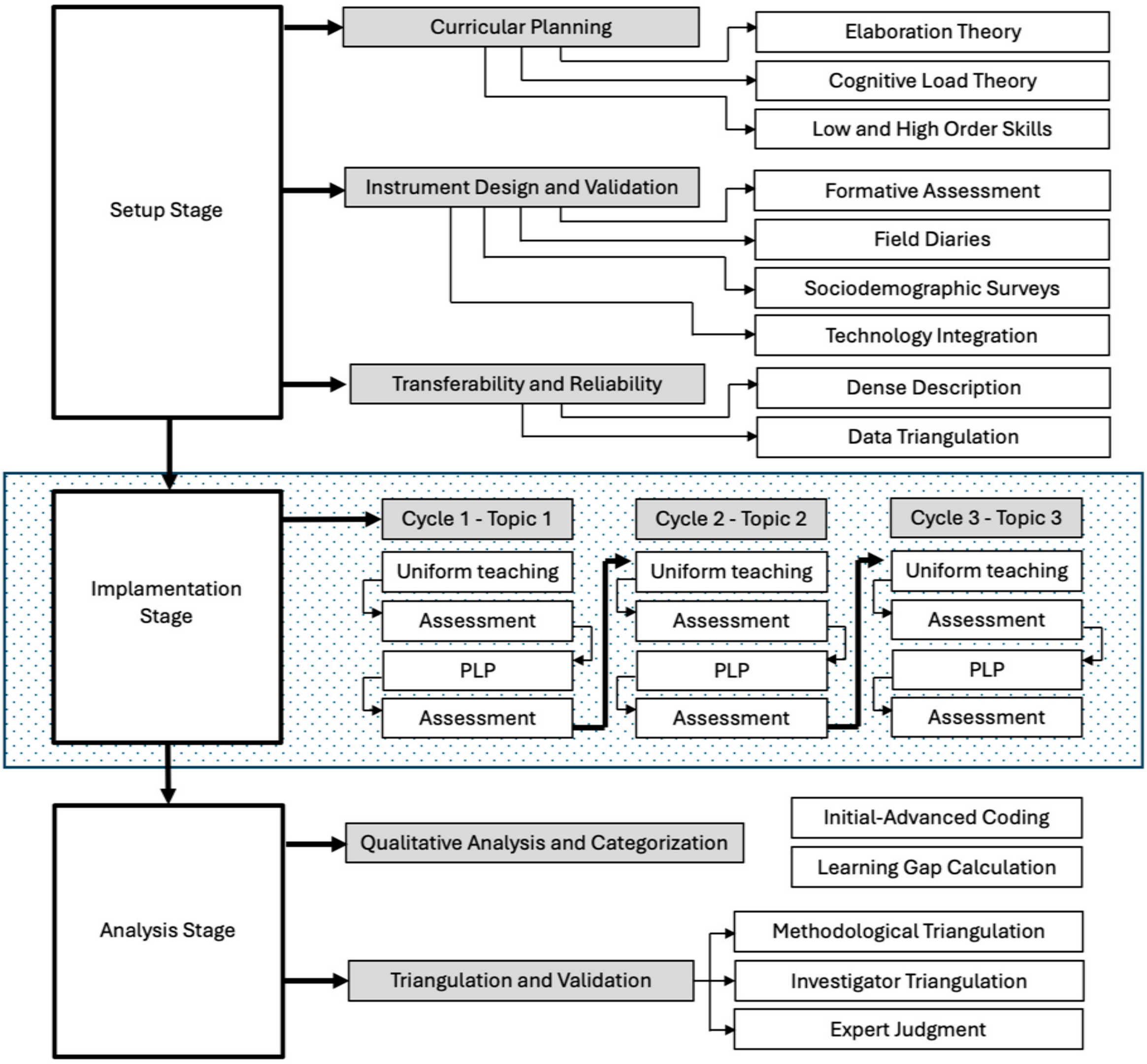


Figure 2. Research procedure.

Instrument Design and Validation

To ensure rigor and validity, four data collection instruments were designed based on established methodological principles [46,47,49,50]. First, formative assessments were developed to identify declarative knowledge gaps through multiple-choice questions, open-ended responses, and recall tasks. These assessments allowed for a detailed evaluation of students' knowledge retention and comprehension [51,52]. In parallel, field diaries were employed to document real-time observations of student behaviors, difficulties, and learning patterns, providing valuable qualitative insights into their educational progress [50].

Additionally, a sociodemographic survey was conducted to characterize participants in terms of socioeconomic status, access to technological resources, and educational support at home. This information was crucial in contextualizing the findings and understanding external factors influencing student learning.

Furthermore, the study integrated GenAI-generated PLPs as part of the instructional strategy to support students' autonomous practice between formative assessments. Gemini 2.5 Pro was used as the primary tool because it was freely accessible and operationally stable for students in the participating school, while ChatGPT 5 was employed only as a contingency option when account or compatibility constraints prevented access to Gemini 2.5 Pro. To minimise variability in outputs, a standard prompt template was used to capture each student's learning gaps, target Bloom level(s), preferred learning-style cues, and required response format. In addition, teacher-facing guidelines were established to verify alignment with intended curricular outcomes and to correct or redirect outputs when necessary. Because the study was not designed as a tool-comparison experiment and tool use was not balanced across students or cycles, results are reported at the level of GenAI-supported PLPs rather than attributing effects to a specific model.

Strategies for Improving Transferability and External Reliability

External consistency refers to the ability to apply research findings to similar contexts, ensuring the broader relevance of the study. To enhance transferability and external reliability, three key strategies were implemented [50,53]. First, a dense description and contextualization approach was employed to provide detailed documentation of the educational setting, participant characteristics, and the specific conditions under which data collection instruments were applied. This comprehensive account ensured a clearer understanding of the study's scope and applicability in different educational contexts.

Additionally, data triangulation was conducted by cross-comparing findings from formative assessments and researcher field notes. This process strengthened the coherence and validity of the results, ensuring that multiple data sources supported the identified patterns and conclusions. Lastly, expert judgment was incorporated to evaluate the quality of formative assessment items and cognitive process categories. Experienced educators selected based on their disciplinary knowledge, pedagogical expertise, and academic qualifications—requiring at least a master's degree—assessed the study's instruments following the criteria outlined by Escobar-Pérez and Cuervo-Martínez [54]. Their evaluation ensured that the assessments were clear, coherent, and methodologically sound, contributing to the overall reliability of the study.

4.3.2. Implementation Stage

The research design enabled the observation of participant progress over time, assessing their performance through three iterative cycles, each consisting of four distinct components. In the first stage, students engaged with a specific subtopic within a unit through three lessons designed and delivered using a standardized instructional approach. This homogeneous teaching strategy incorporated explicit and direct instruction applied equally to all students, integrating problem-based learning and experiential learning techniques to foster conceptual understanding.

Following this instructional phase, the second stage involved the administration of a formative assessment to identify potential learning gaps. These assessments were analyzed using content analysis techniques, categorizing student responses into knowledge gaps, comprehension, application, analysis, and evaluation [46]. The data were systematically coded and stored in a structured Excel database, allowing for a frequency analysis of learning gaps. Based on the assessment results, students received targeted feedback, aimed at reinforcing learning areas where gaps were identified.

In the third stage, students developed another subtopic from the same unit through a Personal Learning Pathway (PLP) tailored to their individual needs. These pathways were generated using Generative AI models (Gemini 2.5 Pro or ChatGPT 5) and were customized based on each student's identified learning style preferences (determined through the CHAEA questionnaire) and qualitative teacher-based profiling.

Finally, in the fourth stage, a second formative assessment was conducted to reevaluate learning outcomes and detect any remaining knowledge gaps. Students once again received personalized feedback, reinforcing concepts that required further consolidation and ensuring a more adaptive learning progression.

4.3.3. Analysis Stage

Qualitative Analysis and Categorization

The analysis followed a longitudinal logic in which each student's evidence of learning was interpreted relative to their own prior performance across the study cycles. The primary unit of analysis was the individual student response to each formative-assessment item, complemented by explanatory traces captured in students' written work and the teacher's field diaries. This structure enabled the identification of learning gaps as observable shortcomings in knowledge, reasoning steps, or task execution, rather than as general achievement differences between students.

Qualitative analysis was conducted through a staged coding process. First, student responses were examined to identify recurring misconceptions, missing conceptual links, procedural errors, or incomplete reasoning. These initial codes were iteratively refined into focused categories that represented the nature of the gap (e.g., conceptual misunderstanding, procedural lapse, incomplete justification), while preserving close alignment with the assessed content and the intended learning outcomes. To ensure coherence across cycles, the coding rules were applied consistently to comparable tasks and were updated only when new gap patterns emerged that required clearer category definitions.

Finally, the qualitative categories were integrated with a Bloom-aligned framework to quantify learning gaps by cognitive level. Each coded response was mapped to the cognitive demand of the corresponding item (knowledge, comprehension, application, analysis, synthesis, and evaluation), allowing the study to summarise how gaps changed across cycles not only in frequency but also in cognitive complexity. This produced cycle-by-cycle indicators of lower-order versus higher-order gap reduction, which were then contrasted between homogeneous instruction and GenAI-supported PLPs to examine patterns of progress under each instructional approach.

In the advanced coding phase, initial codes were refined to form broader analytical categories. This allowed for a deeper understanding of how instructional strategies influenced learning, particularly comparing homogeneous teaching to personalized learning pathways (PLPs). Patterns of improvement, stagnation, and decline were identified, enriching the analysis of learning trajectories across the study's cycles.

The final analytical step involved quantifying learning gaps using Bloom's Taxonomy. Each student's performance was classified according to cognitive levels—knowledge, comprehension, application, analysis, synthesis, and evaluation—providing clear insights into areas of difficulty. Cross-referencing data from formative assessments, teacher observations, and AI-driven learning interactions allowed for a more precise mapping of academic deficiencies. This multi-source validation strengthened the reliability of the findings, offering a robust view of individual learning processes and the effectiveness of personalized instructional approaches.

Triangulation and Validation

To strengthen the credibility of the findings, the study relied on triangulation as a principle for interpreting learning-gap change through converging evidence. Specifically, learning-gap classifications derived from formative-assessment responses were cross-checked against complementary sources, including the teacher's field diaries and students' learning artefacts generated during and between cycles. This approach reduced the likelihood that conclusions would depend on a single data stream and supported more robust interpretations of change over time.

Investigator triangulation was used to minimise the risk of subjective interpretations influencing the analysis. The researchers compared coding decisions and interpretations and refined category definitions when ambiguities emerged, prioritising consistent application of the coding rules across cycles. When interpretations diverged, they were resolved through analytic discussion focused on the observable evidence in student responses and the alignment with intended learning outcomes.

In addition, expert judgement served as a validation strategy for the assessment instruments and their alignment with the intended curricular objectives. Experienced educators reviewed the formative assessments and verified that the items appropriately represented the targeted content and cognitive demands. This expert review supported the adequacy of the instruments for detecting learning gaps and for tracking change across the cycles.

Additionally, investigator triangulation was employed to minimize potential researcher bias. This process involved multiple researchers participating in the coding and interpretation of data, facilitating a broader perspective on learning patterns and knowledge gaps. The engagement of multiple analysts ensured a more objective and balanced assessment, reducing the risk of subjective interpretations influencing the conclusions.

Furthermore, expert judgment was incorporated as a validation strategy to reinforce the methodological rigor of the study. A panel of experienced educators conducted a thorough evaluation of assessment instruments, ensuring that each item met the criteria of clarity, coherence, and relevance as established by Excobar-Pérez and Cuervo-Martínez [54]. Experts were selected based on their academic qualifications, disciplinary expertise, and professional teaching experience, guaranteeing that their evaluations were aligned with best practices in educational assessment. This external validation process contributed to the refinement of assessment tools, ensuring that they effectively captured the intended learning outcomes.

4.4. Ethical Considerations

The study adhered to strict ethical research principles to ensure the protection, confidentiality, and well-being of all participants. A fundamental aspect of ethical compliance was confidentiality, where student data were fully anonymized, and research records were securely stored to prevent unauthorized access. This measure safeguarded participants' personal information and ensured compliance with data protection regulations.

Furthermore, informed consent was obtained from all participants and their legal guardians prior to their involvement in the study. A clear and transparent explanation of the study's objectives, methodologies, potential risks, and expected outcomes was provided to ensure that participation was fully voluntary. This consent process emphasized the right of participants to withdraw from the study at any time without consequences, reinforcing ethical research standards.

Additionally, the study underwent institutional review and ethical clearance from the relevant academic committees where applicable. This approval process ensured that all research procedures aligned with ethical guidelines for educational research, guaranteeing that participants were treated with fairness, respect, and protection throughout the study.

5. Results

This section reports the findings in two steps. First, we describe students' preferred learning-style profiles as a contextual descriptor for how PLPs were configured. Second, we report learning outcomes across three monthly cycles, contrasting homogeneous instruction with GenAI-supported Personalised Learning Pathways (PLPs). For each cycle, we summarise overall performance, changes in lower-order learning gaps, and changes in higher-order learning gaps, using the corresponding figures to display distributions and limiting the narrative to the most salient patterns.

The data analysis examined the evolution of learning gaps by comparing a personalized learning pathway (PLP) with homogeneous teaching-learning strategies, such as explicit and direct instruction, problem-based learning, and hands-on experimentation. The comparison considered units with different levels of relevant cognitive load, determined by the level of cognitive progression and the prior knowledge required. This section presents the following results: predominant learning style profiles and student performance in relation to the relevant cognitive load of each unit, analyzing the effectiveness of PLPs in reducing learning gaps.

5.1. Results on Preferred Learning Style Profiles

The analysis of predominant learning styles indicates that participants do not follow a single fixed pattern, but rather transition between different styles (Active, Theoretical, Pragmatic, and Reflective), depending on their individual characteristics. This finding aligns with the idea that learning is not static but rather a process influenced by multiple factors. However, clear tendencies emerged in the preference for certain styles. In this study, students exhibited combinations of two learning styles (Active and Reflective), as determined by the highest scores obtained in the CHAEA questionnaire, as illustrated in Figure 3.

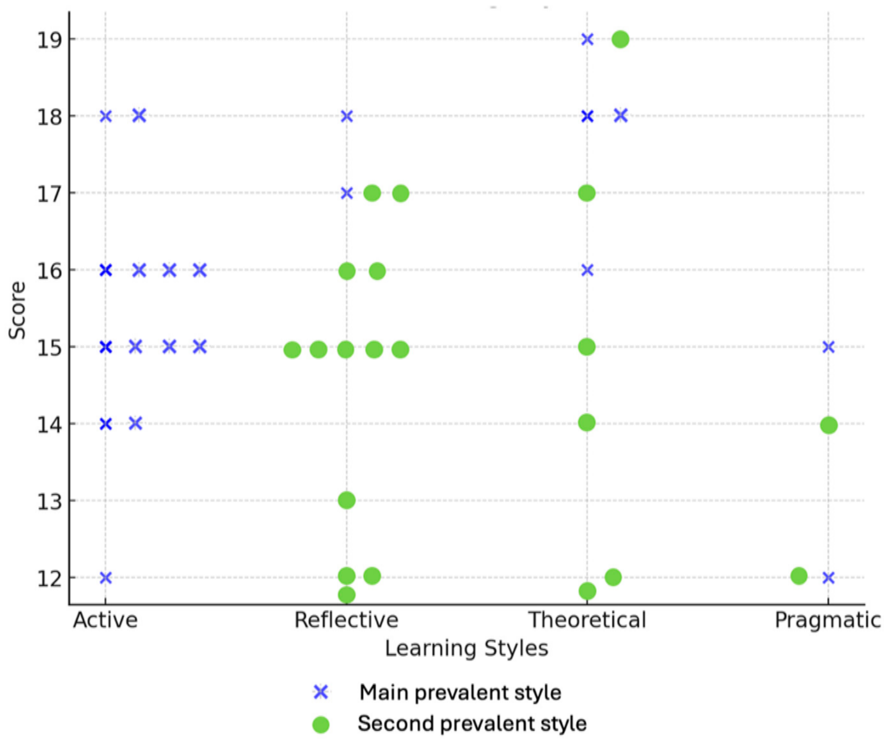


Figure 3. Scatter Plot of Learning Styles and Scores.

These findings reinforce the notion that individuals develop specific learning inclinations, which facilitate both the acquisition of new knowledge and the consolidation of prior concepts within their cognitive structure. Honey and Mumford argue that learning, in addition to being experiential, is influenced by attitudinal and behavioral traits [14,15]. In this sense, the combination of learning styles with individual characteristics enables Generative AI (GAI) to personalize instruction, taking these differences into account. Based on the data obtained through the CHAEA instrument, a GAI-supported PLP was generated for each student in every personalized learning session, offering precise and scalable adaptations.

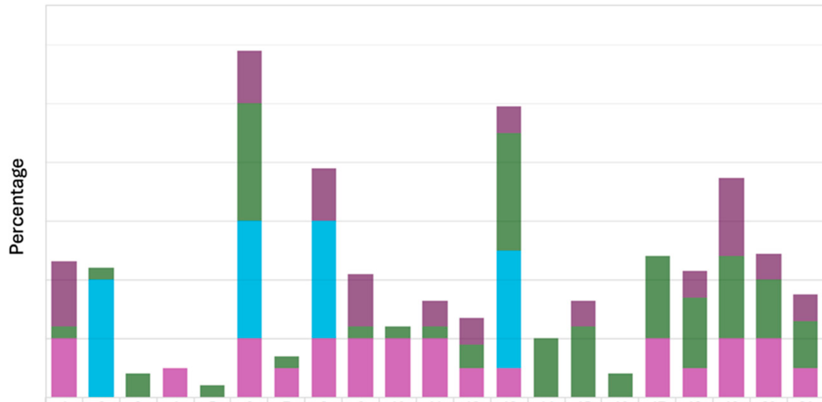
Overall, the profiles show variability across the four CHAEA dimensions, with some students presenting balanced preferences and others exhibiting clearer predominance patterns. These distributions provide the descriptive basis for configuring the PLPs and for examining whether preferred styles relate to how students engaged with the AI-generated explanations across the cycles.

5.2. Student Performance Results

Student performance is reported across three monthly instructional cycles designed with increasing cognitive progression. In each cycle, students completed a formative assessment aligned with the targeted content and Bloom-related cognitive demands, and learning gaps were identified through qualitative analysis of student responses and summarised by cognitive level. We report results by cycle and by instructional strategy (homogeneous versus GenAI-supported PLPs), focusing on the distribution of learning gaps rather than repeating the information already displayed in the figures. Where illustrative cases are provided, they are used only to exemplify the most representative patterns observed.

5.2.1. First Cycle: Basic Cognitive Progression Level

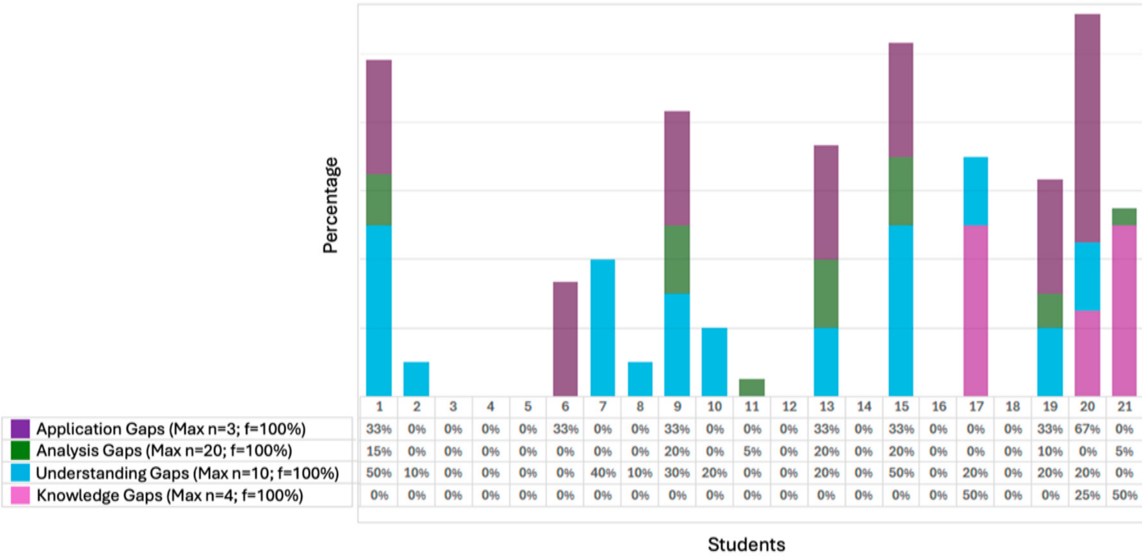
Cycle 1 examined students' initial consolidation of foundational content under two instructional approaches. Under the homogeneous strategy, students achieved formative-assessment scores ranging from 63% and 96%. (n = 18 meeting the passing threshold), whereas under GenAI-supported PLPs, scores ranged from 73% to 100% (n = 21 meeting the passing threshold). Figures 4 and 5 display the distribution of learning gaps under each approach. Overall, the cycle suggests that both strategies supported basic progression, while gap reduction patterns differed in magnitude and in the types of remaining difficulties.



	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Application Gaps (Max n=9; f=100%)	56%	0%	0%	0%	0%	44%	0%	44%	44%	0%	22%	22%	22%	0%	22%	0%	0%	22%	67%	22%	22%
Analysis Gaps (Max n=10; f=100%)	10%	10%	20%	0%	10%	100%	10%	0%	10%	10%	10%	20%	100%	50%	60%	20%	70%	60%	70%	50%	40%
Understanding Gaps (Max n=4; f=100%)	0%	100%	0%	0%	0%	100%	0%	100%	0%	0%	0%	0%	100%	0%	0%	0%	0%	0%	0%	0%	0%
Knowledge Gaps (Max n=4; f=100%)	50%	0%	0%	25%	0%	50%	25%	50%	50%	50%	50%	25%	25%	0%	0%	0%	50%	25%	50%	50%	25%

Figure 4. Learning Gaps—Cycle 1—Homogeneous Teaching.

A qualitative analysis, triangulated with field diaries, identified seven students (33.3%) who demonstrated optimal performance in the personalized strategy, with no gaps in the cognitive categories assessed. Notably, students 3, 15, and 16 exhibited higher-order skill improvements (synthesis). The factors contributing to success included peer collaboration, self-regulated learning, strategic AI interaction, and active engagement with concrete materials.



	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Application Gaps (Max n=3; f=100%)	33%	0%	0%	0%	0%	33%	0%	0%	33%	0%	0%	0%	33%	0%	33%	0%	0%	0%	33%	67%	0%
Analysis Gaps (Max n=20; f=100%)	15%	0%	0%	0%	0%	0%	0%	0%	20%	0%	5%	0%	20%	0%	20%	0%	0%	0%	10%	0%	5%
Understanding Gaps (Max n=10; f=100%)	50%	10%	0%	0%	0%	0%	40%	10%	30%	20%	0%	0%	20%	0%	50%	0%	20%	0%	20%	20%	0%
Knowledge Gaps (Max n=4; f=100%)	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	50%	0%	0%	25%	50%

Figure 5. Learning Gaps—Cycle 1—Personalized Learning Pathways.

First Cycle Lower-Order Learning Gaps

The formative assessment helped identify students who significantly reduced learning gaps in lower-order cognitive skills through personalized learning. Six students (28.5%) achieved positive results, demonstrating that PLPs strengthened basic cognitive skills.

The field diary analysis suggests that improvements were linked to personalized resources aligned with students' learning styles and interests. For example, student 6 (Pragmatic–Reflective, low initial participation) was frequently distracted in the homogeneous strategy, but in the personalized strategy, he showed increased motivation and engagement, especially when interacting with Gemini 2.5 Pro AI. The immediate feedback, structured test interactions, and tailored responses helped clarify concepts and enhance knowledge retention. Additionally, he engaged in autonomous learning outside class, presenting evidence of progress via mobile devices and participating in closing sessions.

Similarly, student 13 (Theoretical–Reflective, preference for diverse resources) improved due to peer collaboration and varied materials suited to her learning style. She relied on teacher guidance, discussions with peers, AI-generated explanations, and interactive games, which enhanced her understanding and confidence. However, eight students (38.1%) showed partial progress, improving in some categories while regressing in others. Cognitive overload was a key challenge, particularly for students 9, 17, 19, 20, and 21, who found Gemini's 2.5 Pro reading materials excessive, despite later acknowledging their clarity and relevance.

Additionally, students with a predominant Active learning style found it difficult to work independently, preferring group-based tasks. Student 7 (Active–Reflective, social difficulties) struggled to seek support, impacting his comprehension performance. Similarly, student 20 (Active–Reflective, low concentration) required stricter supervision and self-regulation strategies due to frequent distractions, including social media use.

First Cycle Higher-Order Learning Gaps

The formative assessment focused on mastery learning, identifying students who significantly reduced gaps in higher-order skills such as analysis through personalized learning. Besides those who had no prior difficulties, notable improvements were observed in students 6, 13, 17, 19, and 21, who initially had substantial learning gaps in the homogeneous strategy but improved significantly under personalized instruction. Additionally, students 2, 7, 11, 15, and 20 managed to reduce their learning gaps, while student 8 maintained stable performance.

A key finding is that 85.7% of students improved their ability to analyze information, identify relationships, and recognize patterns, which are critical 21st-century skills linked to problem-solving, decision-making, and critical thinking. The data suggest that these improvements stemmed from prior reinforcement of lower-order skills, as higher-order thinking relies on a strong foundational knowledge base.

However, students 1, 9, and 10 showed a slight increase in learning gaps related to analysis, which was correlated with lower comprehension performance. These findings confirm the interdependence between cognitive processing levels, reinforcing that higher-order skills require a solid mastery of basic cognitive abilities.

5.2.2. Second Cycle Results: Intermediate Cognitive Progression Level

Cycle 2 assessed student progress at an intermediate cognitive progression level and documented how learning gaps evolved after the first cycle. Under the homogeneous strategy, students obtained scores between 65% and 92% (n = 19 meeting the passing threshold), while under GenAI-supported PLPs, scores ranged between 78% and 100% (n = 21 meeting the passing threshold). Figures 6 and 7 summarise the learning-gap distributions for each approach. At this stage, gap reduction remained more consistent for lower-order demands, while higher-order gaps showed greater variability across learners.

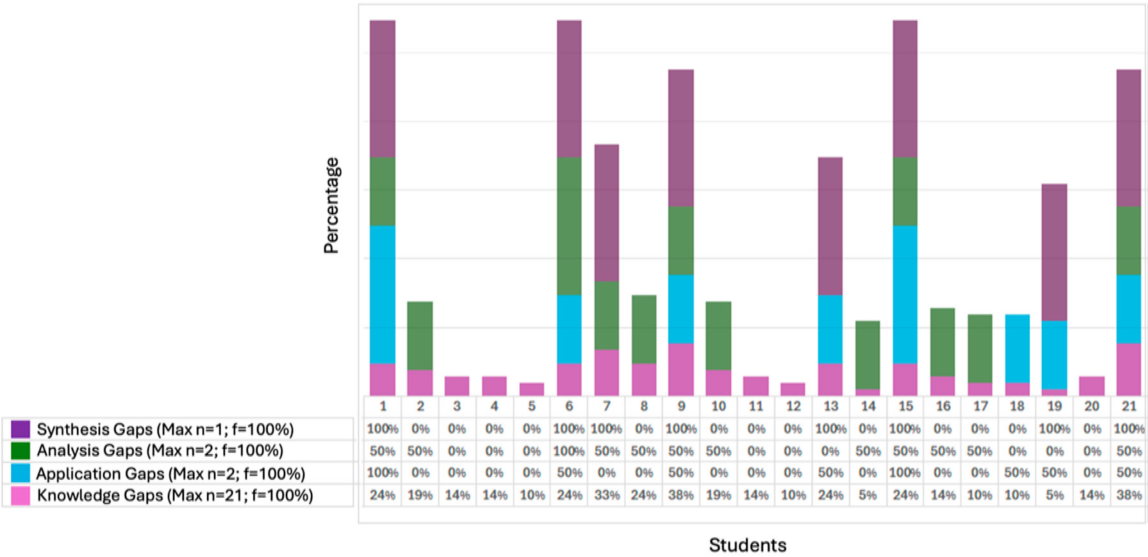
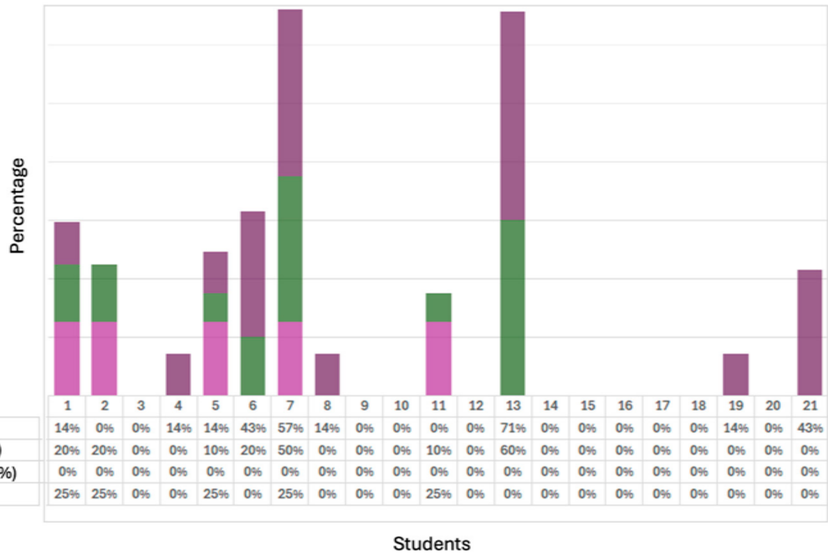


Figure 6. Learning Gaps—Cycle 2—Homogeneous Teaching.

A qualitative analysis, supported by field diaries, confirmed better performance in the personalized strategy, as ten students (47.6%) achieved optimal results (students 3, 9, 10, 12, 14, 15, 16, 17, 18, and 20). Notably, students 3, 12, 14, 16, and 18 maintained sustained progress across cycles. Among them, five students had an Active–Reflective learning style according to Honey & Mumford, potentially benefiting from the alignment between PLP resources and their learning preferences. The remaining five students, with varied learning styles, also improved, indicating that PLPs effectively adapted to diverse learner needs.

Key factors contributing to success included strategic AI interaction, engagement with tailored resources, improved organization, self-regulation, and peer collaboration. In particular, students 3, 14, 15, 16, 18, and 20 leveraged concrete materials, enhancing their understanding and application of the topic.



	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Analysis Gaps (Max n=7; f=100%)	14%	0%	0%	14%	14%	43%	57%	14%	0%	0%	0%	0%	71%	0%	0%	0%	0%	0%	14%	0%	43%
Application Gaps (Max n=10; f=100%)	20%	20%	0%	0%	10%	20%	50%	0%	0%	0%	10%	0%	60%	0%	0%	0%	0%	0%	0%	0%	0%
Understanding Gaps (Max n=2; f=100%)	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Knowledge Gaps (Max n=4; f=100%)	25%	25%	0%	0%	25%	0%	25%	0%	0%	0%	25%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%

Figure 7. Learning Gaps—Cycle 2—Personalized Learning Pathways.

Second Cycle Lower-Order Learning Gaps

The formative assessment identified students who significantly reduced gaps in lower-order cognitive skills through personalized learning. Seven additional students (33.3%) achieved positive results, reinforcing the effectiveness of PLPs in strengthening basic skills. Field diary analysis suggests that improvements were linked to greater confidence in AI tools, better executive function skills, and increased engagement during in-class and home study sessions.

Student 4 (Reflective–Theoretical, high academic performance) excelled in Cycle 1 but faced challenges differentiating voltage behavior in series and parallel circuits. Despite this, he maintained a strong overall performance, with feedback helping address this gap.

Student 6 (Pragmatic–Reflective, low initial participation) showed increased engagement and autonomy, preferring to complete PLPs at home, even using AI for other subjects. Student 21 (Active–Reflective, slow work pace) struggled with timely task completion in the homogeneous strategy but demonstrated greater focus and motivation with PLPs, extending the methodology to other subjects.

Conversely, student 13 (Theoretical–Reflective, reluctance to use concrete materials) improved in knowledge acquisition but regressed in application, struggling with virtual circuit simulations due to difficulties positioning measurement tools. Despite teacher support, her hesitation to use hands-on materials resulted in a partial impact from personalization.

Additionally, students 2 and 5 performed better under the homogeneous strategy. Student 2 (Active–Reflective, high academic achievement) experienced declines in knowledge (−6%) and application (−20%) with PLPs, possibly due to insufficient AI guidance for instrument placement in simulations and lack of teacher support. Student 5 (Reflective–Theoretical, high academic achievement) also had lower knowledge (−15%) and application (−10%) scores with PLPs, struggling to compare series and parallel circuits without peer collaboration, which was more available in homogeneous instruction.

Second Cycle Higher-Order Learning Gaps

The formative assessment identified students who significantly reduced gaps in higher-order skills (analysis) through personalized learning. Students 1, 2, 6, 8, and 21, who initially exhibited significant gaps under homogeneous instruction, demonstrated substantial improvement with PLPs. Student 11 maintained stable performance.

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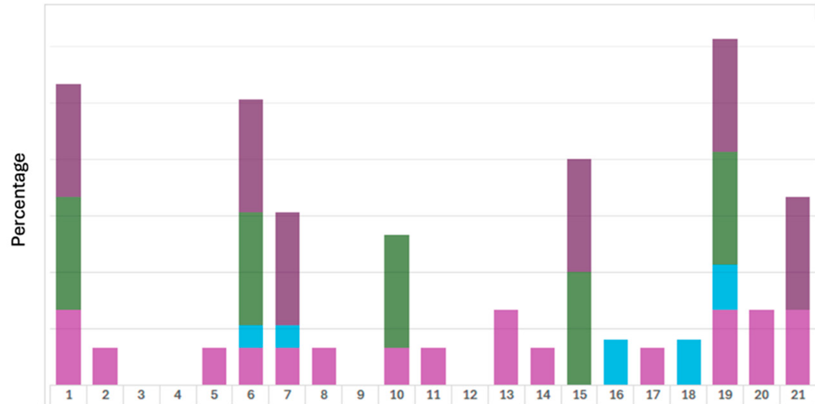
A notable 76.2% of students successfully engaged in AI-guided experimentation, analyzing circuits through interpretation, synthesis, and real-world application. This allowed them to differentiate voltage, current, and resistance behaviors in series and parallel circuits, enhancing their critical thinking and conceptual transferability.

However, students 4, 5, 13, and 19 showed slight increases in analysis gaps, possibly linked to lower comprehension or application performance. Student 19 (Active–Theoretical, easily distracted in discussions) demonstrated strong analysis skills but weaker synthesis skills under homogeneous instruction. While her PLP experience improved analytical performance (only a 14% analysis gap remained), it was insufficient for full higher-order skill consolidation. Overall, findings confirm the interdependence between cognitive processing levels, as higher-order skills depend on strong foundational knowledge and comprehension.

5.2.3. Third Cycle Results: Advanced Cognitive Progression Level

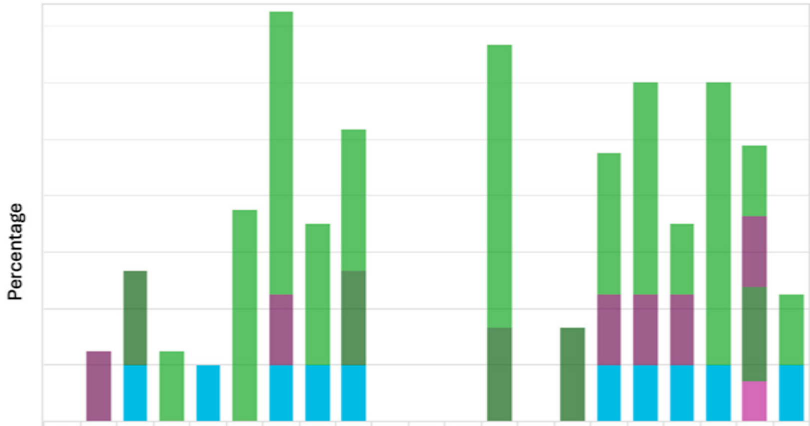
Findings suggest that both instructional strategies are effective and complementary, though personalized learning proves more effective in reducing learning gaps. In this cycle, where cognitive load was significantly higher, most students successfully integrated new knowledge with preexisting structures, supporting the idea that PLPs serve as scaffolding aligned with Vygotsky's zone of proximal development.In the homogeneous strategy, 20 students scored between 60% and 100% in declarative knowledge, yet gaps persisted in comparing resistance and current variations when voltage remains constant and applying Ohm's Law to resistance calculations. Reinforcement strategies included feedback, concrete materials, and simulations (Phet Colorado, Crocodile, and Tinkercad). Figures 8 and 9 illustrate the learning gap distribution in both strategies.



	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Evaluation Gaps (Max n=1; f=100%)	100%	0%	0%	0%	0%	100%	100%	0%	0%	0%	0%	0%	0%	0%	100%	0%	0%	0%	100%	0%	100%
Analysis Gaps (Max n=1; f=100%)	100%	0%	0%	0%	0%	100%	0%	0%	0%	100%	0%	0%	0%	0%	100%	0%	0%	0%	100%	0%	0%
Application Gaps (Max n=5; f=100%)	0%	0%	0%	0%	0%	20%	20%	0%	0%	0%	0%	0%	0%	0%	0%	40%	0%	40%	40%	0%	0%
Understanding Gaps (Max n=3; f=100%)	67%	33%	0%	0%	33%	33%	33%	33%	0%	33%	33%	0%	67%	33%	0%	0%	33%	0%	67%	67%	67%

Figure 8. Learning Gaps—Cycle 3—Homogeneous Teaching.

Cycle 3 focused on an advanced cognitive progression level, where tasks demanded increased integration and reasoning. Under the homogeneous strategy, students achieved scores between 60% and 100% (n = 20 meeting the passing threshold), whereas under GenAI-supported PLPs, the personalized strategy effectively reduced lower-order cognitive gaps, some students still struggled with higher-order skills, particularly synthesis, indicating the need for further learning experiences. Figures 8 and 9 show the corresponding distributions of learning gaps. Compared with earlier cycles, lower-order gaps were reduced more consistently than higher-order gaps, indicating that advanced demands introduced greater dispersion in students' learning trajectories.



Students	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Synthesis Gaps (Max n=4; f=100%)	0%	0%	0%	25%	0%	75%	100%	50%	50%	0%	0%	0%	100%	0%	0%	50%	75%	25%	100%	25%	25%
Analysis Gaps (Max n=; f=100%)	0%	25%	0%	0%	0%	0%	25%	0%	0%	0%	0%	0%	0%	0%	0%	25%	25%	25%	0%	25%	0%
Application Gaps (Max n=3; f=100%)	0%	0%	33%	0%	0%	0%	0%	0%	33%	0%	0%	0%	33%	0%	33%	0%	0%	0%	0%	33%	0%
Understanding Gaps (Max n=5; f=100%)	0%	0%	20%	0%	20%	0%	20%	20%	20%	0%	0%	0%	0%	0%	0%	20%	20%	20%	20%	0%	20%
Knowledge Gaps (Max n=7; f=100%)	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	14%	0%

Students

Figure 9. Learning Gaps—Cycle 3—Personalized Learning Pathways.

In the personalized strategy, 21 students improved their learning with less variability, achieving scores between 74% and 100%. The qualitative analysis of formative assessments and field diaries confirmed better performance with PLPs, as five students (28.6%) showed optimal performance (students 1, 6, 10, 11, 12, and 14). Notably, students 10, 12, and 14 demonstrated sustained progress compared to previous cycles. Half of the top-performing students had an Active–Reflective learning style, which aligns with the group's predominant learning preference. These students successfully connected new knowledge with prior concepts, reinforcing long-term retention through progressive learning cycles and targeted strategies that addressed weaknesses while celebrating achievements [55].

The remaining three high-achieving students had different learning styles, confirming that PLPs effectively adapt to diverse learner needs. Key success factors included increasingly strategic AI interaction, autonomous simulation-based exercises, unlimited AI-generated explanations, and examples tailored to individual needs. These students required minimal teacher support.

Additionally, students developed executive functions such as organization, time management, and planning, which proved crucial for academic success. Peer collaboration also played a fundamental role, as students shared both technical and disciplinary knowledge, resolving doubts through discussion and debate.

Third Cycle Lower-Order Learning Gaps

The formative assessment identified students who significantly reduced gaps in lower-order cognitive skills through personalized learning. Ten additional students (47.8%) demonstrated progress, highlighting the effectiveness of PLPs in reinforcing foundational skills. Field diary analysis suggests that improvements were driven by high student engagement, motivation, and positive AI interaction. Flexible pacing, additional school learning spaces, and strong alignment between AI-generated resources and individual learning profiles contributed to their success.

Notably, students 1, 6, 7, 13, 19, and 20, who previously struggled with low participation, frequent distractions, and group work disengagement, performed significantly better in PLPs, receiving fewer disciplinary interventions. The PLP structure enhanced focus, confidence, and knowledge retention, ultimately improving both academic and behavioral performance.

Conversely, students 15, 16, and 18 (14.3%) showed partial progress, improving in some areas while regressing in others. Their difficulties stemmed from AI limitations in providing step-by-step mathematical operations, particularly when applying Ohm's Law. Although these students had strong logical and mathematical reasoning skills, they preferred teacher-led explanations over AI guidance, citing concerns about AI accuracy and the high cognitive load of the unit.

Similarly, students 3 and 9 performed better in the homogeneous strategy. Student 3 (Active–Theoretical, high academic achievement) struggled with comprehension (−20%) and application (−33%) in PLPs, possibly because AI-generated explanations lacked interpretative depth beyond direct instruction.

Student 9 (Active–Reflective, easily distracted) also had lower comprehension (−20%) and application (−33%) scores in PLPs. His difficulties with mathematical operations and algebraic expressions hindered his performance. However, once foundational knowledge was reinforced, he showed greater engagement and even achieved 100% accuracy in the homogeneous strategy's formative assessment.

Third Cycle Higher-Order Learning Gaps

The formative assessment identified students who significantly reduced gaps in higher-order skills (analysis and synthesis) through personalized learning. Students 2, 4, 15, 19, and 21, who initially had significant gaps in the homogeneous strategy, showed substantial improvement under PLPs. Student 5 maintained stable performance.

Compared to previous cycles, fewer students demonstrated strong higher-order cognitive skills (analysis and synthesis) in an advanced progression level under PLPs. Only 47.6% successfully developed an enhanced solution to a real-world problem using their accumulated knowledge, applying critical thinking to justify the originality and feasibility of their idea. These skills are essential for lifelong learning, design thinking, and 21st-century problem-solving.

Students 4, 5, and 21 benefited equally from both strategies, exhibiting strong higher-order cognitive abilities regardless of the instructional approach. Their results suggest that students with solid cognitive foundations can develop advanced reasoning skills regardless of instructional strategy when cognitive load is high. These findings confirm the interdependence between cognitive processing levels, as higher-order thinking relies on strong foundational skills.

In contrast, students 3, 7, 8, 9, 13, 16, 17, 18, and 20 (42.8%) struggled to achieve higher-order skills under PLPs. While PLPs offer challenging problem-solving tasks, their resources were insufficient for consolidating complex reasoning processes when cognitive load increased. Notably, the Active–Reflective learning style predominated (5 of 9), aligning with the overall group trend. These findings highlight the need to complement personalized learning with active strategies that foster critical thinking and collaborative problem-solving.

Integrating case studies, structured debates, design thinking sessions, creativity workshops, prototyping activities, and guided solution development could enhance real-world knowledge application. This hybrid approach would leverage PLP flexibility while incorporating the cognitive benefits of peer collaboration and teacher mediation, ultimately fostering stronger, transferable higher-order cognitive skills.

6. Discussion

This study examined the cyclical implementation of GenAI-supported Personalised Learning Pathways (PLPs) to mitigate learning gaps in secondary education. Across the three cycles, the findings point to a stable pattern: PLPs were associated with more reliable consolidation of lower-order learning (knowledge, comprehension, and application), whereas progress on higher-order demands (analysis, synthesis, and evaluation) remained more variable as cognitive complexity increased. Importantly, these patterns emerged in a low-income public-school context where access to devices and connectivity was not uniform, indicating that observed benefits are inseparable from the equity conditions under which students can actually engage with personalised support.

6.1. Implications of Findings

The evidence aligns with prior research on AI-driven education and adaptive learning, reinforcing the pedagogical value of personalisation when it is grounded in clear learning targets and iterative feedback loops [2,56]. In this regard, the results suggest that GenAI-supported PLPs can function as classroom-feasible scaffolds that help learners consolidate prerequisites and stabilise foundational understanding, especially when students need structured practice opportunities that are difficult to deliver through homogeneous instruction alone.

A key implication is that PLPs appear to operate as a form of scaffolded progression consistent with Vygotskian perspectives on guided learning, in the sense that support can be adjusted to what learners can accomplish with assistance and progressively faded as competence increases. Moreover, the integration of AI-generated feedback and adaptive explanations can strengthen self-regulated learning by enabling learners to rehearse, verify, and correct their reasoning between formative assessments, which resonates with research on AI's potential to enhance formative assessment practices [10,42]. Nevertheless, the present findings also indicate that the pedagogical contribution of PLPs is not uniform across cognitive demands, which has direct implications for how they should be designed and used in classrooms.

Although lower-order learning gaps were reduced more consistently, higher-order gaps persisted for a subset of students, particularly when tasks required transfer to unfamiliar contexts and the production of coherent justifications. This outcome echoes prior evidence that higher-order reasoning cannot be assumed to follow automatically from improved recall and application, especially when learners must integrate multiple concepts under increased cognitive demand [16]. Therefore, PLPs should be understood as necessary but not sufficient for deep learning: they can consolidate the foundations that make advanced reasoning possible, yet higher-order progress typically requires complementary instructional strategies that deliberately cultivate explanation quality, argumentation, and reflective evaluation. In practice, approaches such as problem-based learning, debate, and collaborative projects remain crucial for engaging students in metacognitive activity and for making reasoning visible and discussable [15,43]. Accordingly, the most defensible interpretation is that PLPs are most effective when embedded within a broader pedagogical design that explicitly targets higher-order thinking, rather than being treated as a standalone solution.

A further implication concerns learner variability and the role of learning-style preferences in AI-supported personalisation. The literature continues to debate whether learning styles constitute an evidence-based construct or a neuromyth, and this controversy requires caution in how style-based cues are used in instructional design [38]. At the same time, applied research has argued that preference-sensitive adjustments can support engagement and perceived relevance for some learners [15]. Within this study, the observed heterogeneity across profiles suggests a pragmatic stance: learning-style information may be treated as a flexible design cue that shapes the form of explanations and resources, while progress evidence from formative assessment remains the primary driver of pathway decisions. This further supports a hybrid human-AI personalisation approach, where generative

AI contributes adaptive scaffolding and teachers provide pedagogical mediation, quality control, and intentional design for higher-order reasoning.

6.1.1. Transferability and Boundary Conditions

The present evidence supports analytic rather than statistical generalisation. Some influences are strongly context-dependent, particularly unequal access to devices/connectivity and the degree of teacher mediation available during and between cycles. Nevertheless, the mechanism-level insight that structured, feedback-rich pathways can consolidate foundational learning is likely transferable to other settings, provided that minimum implementation conditions are met: stable opportunities for students to engage with the pathway during planned study time, alignment checks that constrain outputs to the intended curricular targets, and explicit guidance that prevents over-reliance and encourages learners to explain, justify, and verify. Under these conditions, PLPs may function as classroom-feasible support that complements, rather than replaces, instructional design and teacher-led reasoning practices.

6.1.2. Implications for Practice

For classroom implementation, three implications follow. First, PLPs are likely to be most beneficial when they target clearly specified lower-order gaps and provide short, sequenced steps that help learners rebuild missing prerequisites and avoid cognitive overload. Second, to support higher-order reasoning, PLPs should be coupled with teacher-designed tasks that require explanation quality, such as structured justification prompts, comparison of alternative solutions, and peer discussion, so that AI-generated support becomes a trigger for elaboration rather than passive consumption. Third, equity constraints should be addressed explicitly: where access is uneven, schools may require scheduled on-site engagement periods and shared-device routines to prevent personalisation from becoming an additional source of inequality.

These interpretations should be considered regarding methodological and contextual constraints, including the single-site design, unequal access conditions, and the fact that GenAI tools were employed for operational feasibility rather than as a controlled comparison. The next section therefore specifies limitations that shape the strength and scope of the conclusions.

6.2. Limitations of the Study

This study has several limitations that should be considered when interpreting the findings. First, the research followed a single classroom cohort (n = 21) in one low-income public-school context. Although the design supports analytic generalisation, the results should not be interpreted as population-level estimates, and replication across additional sites and cohorts is required to evaluate robustness under different conditions. Second, the study employed a non-experimental longitudinal panel design in an authentic setting. Consequently, the evidence does not support strong causal inference in the way that randomised controlled designs might. Changes observed across cycles may reflect a combination of instructional exposure, maturation effects, and contextual influences that are difficult to isolate in naturalistic school environments, even when systematic procedures and triangulation are applied.

Third, unequal access to devices and connectivity likely moderated how consistently students could engage with PLPs, particularly between classroom sessions. This equity-related constraint is not merely a contextual detail but a substantive boundary condition: if learners cannot reliably access the pathway, the pedagogical mechanism of iterative practice and feedback may be weakened, and personalisation may inadvertently reproduce unequal opportunities to benefit. Fourth, GenAI tools were used for operational feasibility rather than as a controlled tool-comparison experiment. Gemini 2.5 Pro served as the primary platform, while ChatGPT 5 was used as a contingency option when access constraints arose. Because tool use was not balanced across students or cycles, and because the study did not implement a counterbalanced design, any tool-specific differences in output style, quality, or error patterns cannot be ruled out as influencing engagement and learning trajectories. For this reason, outcomes are interpreted as effects of GenAI-supported PLPs at the level of the instructional approach rather than as attributable to a particular model.

Fifth, learning-style information was incorporated as a personalisation cue, yet the learning-styles construct remains contested in the literature. Therefore, the present study does not treat learning styles as stable learner traits nor as a definitive explanatory variable. Instead, learning-style profiles are interpreted pragmatically as preference indicators that may shape how learners engage with explanations and resources, while acknowledging that other factors, such as prior knowledge, motivation, and self-regulation, may play an equal or greater role. Finally, although learning gaps were mapped to Bloom-aligned cognitive levels to enable summaries of change by complexity, categorisation inevitably simplifies the diversity of student reasoning. Despite efforts to increase transparency through explicit coding procedures and triangulation, some interpretive judgement is inherent in qualitative classification of misconceptions and incomplete reasoning. Future studies could strengthen this component through expanded inter-rater protocols and by comparing alternative operationalisations of learning gaps.

6.3. Future Research Directions

Future studies should extend these findings through designs that strengthen transferability, clarify the role of GenAI tools, and address higher-order learning more directly. Multi-site replications across schools with different socioeconomic profiles would be particularly valuable to test whether the observed pattern, more consistent reduction in lower-order gaps than higher-order gaps, persists under varying access conditions and instructional routines. To enable meaningful comparison, such replications should preserve comparable cycle structures and shared assessment frameworks, while documenting implementation conditions (e.g., device availability, connectivity, and planned time-on-task).

A complementary line of inquiry concerns tool effects. Because the present work used Gemini 2.5 Pro as the primary platform with ChatGPT 5 as a contingency option, future research should implement controlled or counterbalanced exposure to each tool, using shared prompt templates and explicit output-quality criteria. This would allow researchers to estimate whether model behaviour, explanation style, or error patterns systematically influence engagement, perceived cognitive load, and learning-gap reduction.

Further progress is likely to depend on tighter curricular alignment. A promising direction is the development of a curriculum-aligned language model embedded in a dedicated application that constrains outputs to vetted concepts, examples, and progression rules. Such a system could be built through retrieval-augmented generation based on approved curricular materials or through fine-tuning on curriculum-aligned datasets. A subsequent longitudinal evaluation, using a similar cycle-based approach, could assess whether constrained generation reduces variability, limits misleading outputs, and supports more stable gains in higher-order reasoning.

In addition, interventions should be designed to convert PLP support into opportunities for explanation and transfer rather than relying on practice alone. Hybrid designs that combine PLPs with structured classroom routines, guided argumentation, peer critique, and project-based tasks that require synthesis and evaluation, would help test whether higher-order gaps can be reduced more consistently. These evaluations should include measures that capture not only correctness but also explanatory coherence, reasoning depth, and learners' ability to evaluate the quality of AI-generated assistance.

Finally, longer follow-up periods are needed to determine whether short-term gap reduction translates into durable learning and whether repeated engagement with PLPs cultivates stronger self-regulatory habits. Tracking learners beyond a single term would also clarify whether foundational consolidation eventually supports higher-order growth, or whether additional instructional mediation remains necessary across time.

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